

SARAH LIU

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TECHNICAL SKILLS

Certifications: Google AI, Google UX Design, Cisco Networking Basics
Languages: Python, Java, JavaScript, TypeScript, Swift, SQL, C, C++
Backend: Backend Development, Python, Java, FastAPI, Node.js, Express.js, REST APIs, JWT, OAuth, Rate Limiting
Frontend: React, Next.js, TypeScript, SwiftUI, Tailwind CSS, Zustand
Databases: PostgreSQL, SQL, Supabase, MongoDB, Neon, Row Level Security
DevOps & Tools: Git, Docker, Azure DevOps, CI/CD, Vercel, Linux, Bash, Stripe, Software Development Lifecycle
AI Tooling: Groq, GPT-4, Claude Code, Cursor, GitHub Copilot, Codex, Gemini, v0 by Vercel

RELEVANT EXPERIENCE

- ASML-Affiliated Tabletop Cleanroom – Telemetry Dashboard** | [Project ISO](#) Sept 2025 – May 2026
Arduino C++, JavaScript, Web Serial API, Chart.js, Tailwind CSS Fairfield, CT
• Technical lead for a 3-engineer senior design team; designed a dual-microcontroller **Arduino C++** architecture that separated telemetry collection from motion control to reduce sensor polling bottlenecks.
• Wrote firmware aggregating live readings from **10** environmental sensors at **10 Hz** and streamed data over the **Web Serial API** to a **Chart.js** dashboard at under **200ms** end-to-end latency.
• Delivered real-time ISO 14644-1 compliance telemetry used to certify the chamber as ISO Class 6, earning the **Dean's Award for Best Senior Design** and **2nd Place** at the SME Chapter 7 Competition.
- Oat – Productivity & Planning SaaS** | [OatPlanner.com](#) Mar 2026 – Present
Next.js, TypeScript, Supabase, PostgreSQL, Stripe, OAuth, Groq, JWT, Zustand, Vercel Remote
• Sole engineer for a production SaaS with **100+** users; built a **Next.js/TypeScript** planning app with calendar APIs, recurring events, voice scheduling, habit tracking, and **Zustand**-managed client state.
• Designed a custom **JWT** authentication and authorization flow on **Supabase/PostgreSQL**; enforced Row Level Security on user data and added middleware security headers and rate limiting for API routes.
• Integrated Google Calendar **OAuth**, **Stripe** subscriptions, and a **Groq**-hosted LLM behind server-side handlers, keeping billing, scheduling, and AI workflows separated by API boundary.
• Owned the full **software development lifecycle** from schema design and API implementation through deployment on **Vercel**, production fixes, and iterative UX improvements from real user feedback.
- Sportly – Drop-In Sports Event Platform** | In Development June 2026 – Present
Next.js 14, PostgreSQL (Neon), Drizzle ORM, Auth.js v5, Stripe Connect, Resend, Zod, Tailwind CSS, Vercel Remote
• Co-building a sport-agnostic event registration platform in **Next.js 14** App Router with **Stripe Connect** for direct organizer payouts — funds flow to organizers without Sportly ever holding them.
• Designed the **PostgreSQL** schema on Neon with a transactional capacity reservation system to prevent oversell race conditions: spots held for 15 minutes during checkout, released by a **Vercel** cron on expiry.
• Implemented idempotent **Stripe** webhook processing via an insert-first ledger table; built magic-link auth via **Auth.js** v5 with a guest-checkout-to-account claim flow using email as the merge key.
- Heatmapper – Financial Sentiment Backend** | [HeatmapperA](#) May 2026
Python, FastAPI, PostgreSQL, Plotly.js, VADER, Bright Data API, Docker, JWT Remote
• Built a **Python/FastAPI** backend that ingests financial news, Reddit, and Twitter data through **Bright Data** APIs, normalizes records in **PostgreSQL**, and runs **VADER** sentiment analysis for market narrative signals.
• Implemented **JWT** authentication, PBKDF2-SHA256 password hashing, background scanning, and an in-memory TTL cache to control upstream API usage and support concurrent demo sessions.
• Containerized the service with **Docker** and wired REST endpoints to **Plotly.js** heatmaps for options open interest, volume, implied volatility, and price probability views.
- Teaching Assistant – Cloud Computing** Jan 2026 – May 2026
Fairfield University | Azure DevOps, Linux, Bash Fairfield, CT
• Supported **30+** students on **Azure DevOps CI/CD** pipelines, **Linux**, Bash, and containerization labs; identified recurring blockers during grading and helped reduce repeat errors across lab cohorts.

LEADERSHIP

- Team Captain | Fairfield Women's Tennis – NCAA Division I** Sept 2025 – May 2026
Liaison between athletes, coaches, and the athletic department Fairfield, CT
• Primary liaison across athletes, coaching staff, and athletic department for a 10-person D1 roster, managing competing priorities across 25+ hrs/week of practice and travel.
- sliu.log | Technical Content Brand & Community** Sept 2025 – Present
Adobe Creative Suite, Audience Analytics, Short-Form Video Remote
• Grew a technical content brand to **10,000+** followers (3.5% avg. engagement, 2–3× industry benchmark); launched a Chicago-based Discord growing **0** → **400 members in 1 week** off a single viral video.

EDUCATION

- Fairfield University** May 2026
Bachelor of Science in Computer Science, Minor in Mathematics | Honors Program Fairfield, CT
• GPA: 3.80/4.00 | Dean's List 2022–2026 | Upsilon Pi Epsilon Computer Science Honor Society
• Relevant Coursework: Data Structures, Algorithms, Database Systems, Operating Systems, Networks, Object-Oriented Programming